

Metalab: introduction

Society for Arts and Technology [SAT]

June 14, 2022

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1 The [SAT] and its research laboratory, the Metalab

1.1 Society for Arts and Technology [SAT]



Founded in 1996, the Society for Arts and Technology [SAT]¹ is a non-profit organization recognized internationally for its active, leading role in developing immersive technologies, virtual reality and creative use of high-speed networks. With its triple mission as a centre for the arts, training and research, SAT was created to support a new generation of creators/researchers in the digital age.

A gathering place for diverse intelligence, curiosity, knowledge and talent, the Society for Arts and Technology is a live creative laboratory whose unconventional experiences bring together the tangible and the unexpected.

Within its 4 400m² premises one can find the Satosphere², a permanent modular dome, dedicated to the development and presentation of 360° immersive experiences. With a diameter of 18 meters and height of 13 meters, filled with 157 speakers, this unique equipment complements the studios and workshops housed inside the SAT.

The SAT has naturally become a space for pivotal events, where everything is possible, at once stimulating and showcasing trends in technology applied to arts and design.

¹<https://sat.qc.ca/en>

²<https://sat.qc.ca/en/satosphere>

1.2 The Metalab

Established in 2002, the Metalab is the Society for Arts and Technology [SAT]’s research and development lab. The Metalab has a dual mission: to enable the emergence of new immersive experiences, and to make their design and authoring accessible to artists and creators through an ecosystem of free software.

The Metalab’s research fields – telepresence, immersion, video mapping, and spatialized sound – are explored through research and production projects. The Metalab team is comprised of 5 to 10 people with diverse technical expertise including digital audio, computer graphics, networks, software development and multimedia integration. The Metalab welcomes interns in a variety of subjects, even outside the strict scope of the laboratory expertises.

Metalab’s open source software ecosystem aims to cover the entire production of immersive and distributed content: immersive environment editing, video mapping, sound spatialization, group interaction and low latency transmission. They are developed to be scalable over the long term: they support research and partnerships. Within the SAT, these softwares are integrated to the palette of tools of the [SAT] creation residencies and are at the heart of telepresence projects such as those between libraries (Bibliolab ³) and between theaters (Scènes ouvertes ⁴). Our partners include artists, university researchers, schools, designers and industrialists.

1.3 In between art, science and digital creation communities

Metalab is a multidisciplinary research laboratory focusing on group immersion and telepresence in which co-creation, open innovation and knowledge sharing are at the forefront. Its team is currently composed of a dozen researchers specialized in fields ranging from digital music to machine learning, interaction and computer graphics. Its partnerships and research activities are part of an ongoing collaboration with the world of digital and live arts, university research laboratories, digital creation industries, as well as communities gravitating around digital creation.

At the source of the immersive creation device of the [SAT] : since its first years of existence, Metalab has prioritized the know-how and the production of prototypes of computer tools facilitating the implementation of innovative co-creation projects. The

³[urlhttp://sat.qc.ca/en/recherche/bibliolab](http://sat.qc.ca/en/recherche/bibliolab)

⁴[urlhttp://sat.qc.ca/en/scenes-ouvertes](http://sat.qc.ca/en/scenes-ouvertes)

fruits of its research are disseminated via open-source software and scientific communications. These exchanges of knowledge, as well as the pooling of efforts and resources within open communities, have allowed the SAT to develop several departments : the *Labodôme* and *Satellite* host residencies and artistic projects in immersion, while Campus is dedicated to teaching. Today, the [SAT] offers a pathway covering different aspects of immersive creation, and it is not uncommon for creators to navigate among these departments, thus circulating their knowledge and practice.

Today, through its new research residency program ⁵ and its active participation in open communities (hackathon ⁶, meetups, transparent and free software development), Metalab has an environment that makes possible multidisciplinary collaborations that are unique not easily accessible in many other places dedicated to digital arts.

Anchored in scientific research : Metalab's work is initiated by the development of research programs evaluated by a scientific committee. The resulting projects contribute to the research community, notably through the publication of scientific papers and through participation in conferences. These contributions allow Metalab to confront its ideas and results with specialists and current problems, while initiating collaborations with other laboratories, sometimes international, and sometimes with related disciplines.

The scientific research process at Metalab is also stimulated through internships, which remain primarily experimental in nature, particularly MITACS internships.⁷ that allow for the exploration of emerging topics, such as those linking social science issues and Metalab research.

Hand in hand with industry : close to the industries of creation and diffusion of digital arts and creation, Metalab's research has led to large-scale industrial projects, such as the design of the SAT's *Satosphere* in 2011 ⁸ and the Québec network of artistic telepresence *Open Scenes* in 2017 ⁹. That same year, the [SAT] gave birth to the research valorization department at the [SAT], whose first mission was to support the network of more than twenty connected performance venues.

Today, the valorization strategy is on the move and includes taking on pre-commercialization

⁵<https://sat.qc.ca/fr/nouvelles/apel-residence-rd>

⁶<https://sat.qc.ca/fr/nouvelles/hacklab2021>

⁷MITACS funds cutting-edge research projects by creating employment opportunities for graduate students. <https://www.mitacs.ca/fr>

⁸<https://sat.qc.ca/fr/satosphere>

⁹[urlhttps://sat.qc.ca/fr/scenic-telepresence](https://sat.qc.ca/fr/scenic-telepresence)

projects such as the haptic floor. Since 2020, the [SAT] is recognized as a public research center in Québec, thus facilitating its participation in large-scale industrial research projects, possibly through the provision of research services. Moreover, with the recent development of hardware prototypes (Mirador, Audiodice and Scenic light), [SAT] is moving towards a technology transfer strategy based on turnkey and accessible tools. These strategic evolutions are, since 2020, followed by a circle of stakeholders constituted by partners of the [SAT] who accompany the growth of this innovation strategy.

2 Metalab members

Scientific Council

- **François Borrelli** Directeur Général TechnoMontréal et Président-fondateur de OblikMind
- **Jacques de Guise** Directeur du Laboratoire de recherche en imagerie et orthopédie (LIO). ÉTS et CR-CHUM
- **Wisseem Maazoun** Gestionnaire de compte, développement des affaires à MITACS

Direction of Innovation and Training [SAT]

- **Véronique Paradis** is an industrial designer and holds a master's degree in strategy from HEC Montréal. Her leadership in strategic innovation management maximizes the impact and transforms R&D teams through her user-centered, multidisciplinary and experimentation-driven approach. Véronique has led several open innovation projects in the manufacturing industry, notably in the furniture and metal fabrication sectors, introducing companies to design thinking as a strategic lever for differentiation.

Co-direction of Metalab

- **Emmanuel Durand** holds a doctorate in computer graphics from Arts et Métiers ParisTech (2013). He has an interest in everything related to 3D, photography and space.
- **Nicolas Bouillot** has a doctorate in computer science from the Conservatoire National des Arts et Métiers (2005). He likes telepresence, sound spatialization, musical improvisation, and network protocols.

Metalab Team

- **Alexandra Marin** has a master's degree in project management from Université du Québec à Montréal. She is interested in multidisciplinary creation processes and methods for artistic and cultural projects, in digital culture as a vector of social change and in electronic music.
- **Christian Frisson** was a doctoral researcher at the University of Mons (numedi-art institute) and a postdoctoral researcher at Inria (Mjolnir team), at the University of Calgary (Interactions Lab), and at McGill University (IDMIL). He creates tools to author multimedia and immersive multi-user interactions; by computer vision, computer graphics, and haptic and sound feedback.
- **Édith Viau** studied mathematics and has a master's degree in financial engineering. She enjoys the Python language and low-level programming, while being interested in the communities surrounding the free software movement. In her artistic practice, she explores markets as a medium.
- **Edu Meneses** is currently a Ph.D. researcher in music technology at McGill University. Edu works with embedded systems for digital musical instruments and leads research-creation projects with product development and art-oriented usability evaluation. His projects explore gestural control of sound systems, digital signal processing (DSP), mapping (including sound spatialization), instrument/installation design, and music composition/performance.
- **Émile Ouellet-Delorme** has a bachelor's degree in digital music from Université de Montréal (2017) and a bachelor's degree in software engineering from Polytechnique Montréal (2014). He is interested in generative music, 3D, acoustics, and sound spatialization.

- **Farzaneh Askari** is currently a Ph.D. candidate at McGill University, with her research focusing on self-supervised video understanding. She obtained her Master's degree from the University of Denver, working on social robotics for children with Autism. She likes playing badminton, outdoors, tech, and animals, especially cats!
- **Michał Seta** is a composer, improviser and researcher in digital arts. Practitioner of transdisciplinary, transcalar, and integrative magic, he incants Metalab software in collective and improvised harmony.
- **Thomas Piquet** has a doctorate in computer science from the University of Rennes 1 (2008), specializing in embedded systems. A fan of sound, from synthesis to reproduction, he designs his music like images or cooking; the reverse is also true.
- **Zack Settel** has a doctorate in musical composition from the University of Montreal (2003). He is interested in composition with 6 degrees of freedom, involving augmented reality and non-linear musical experiences.

Intern(s)

- **Éliès Jurquet** studies digital technologies for image and multimedia at ENSEEIHT (Toulouse, France). He is currently following a professional master's degree in computer engineering at Polytechnique Montreal. His keen interest in artistic experimentation, VFX and cyberculture has led him today to develop a motion capture tool for Blender at the Metalab, which will allow the manipulation of avatars thanks to LivePose.
- **Emma Laurent** is a PhD candidate in Information and Communication Sciences at the University Paul-Valéry Montpellier 3. Her thesis focuses on gendered and generational representations in the (human and digital) mediation of classical music. Her work with Metalab is within the framework of the project "Questioning the musical experience: contexts, representations and practices", conducted under the supervision of the University of Montreal.
- **Eva Décorps** is studying at École des Mines de Paris. She likes mathematics, computer sciences, arts... And everything that exists at the intersection of these disciplines, she has a soft spot for generative arts.
- **Pierre Gilbert** is an intern at Metalab, working on Audiodice calibration. Music being his childhood passion, he has been nurturing an interest towards acoustics and sound since high school, when he started searching for an university. A musician and saxophonist of 15 years of experience, he is captivated by audio-visual arts. It is the topics of sound and image coordination and immersive audio that brought him to SAT and Metalab.
- **Raphaël Suzor-Schröder** is studying mechanical engineering at McGill University. His greatest dream is to create an amusement park, combining technology, art, and play: an "ARTmusement" park. Being an intern at SAT is hence a good opportunity for him to learn how to connect the different fields and get involved in a community that is passionate about that philosophy. He loves bouldering, ice-skating and rollerblading.
- **Sayaka Araniva-Yanez** is currently pursuing a Bachelor's degree in Literary Studies. Their interests are very much focused on different digital cultures, especially hypertextual literature and electronic music. Digital inclusivity, resistance and sovereignty of minority groups are at the heart of their research.
- **Victor Comby** designs all kinds of sounds with his computer. Always looking for new techniques and experiences, his quest brought him to the Metalab. There

he shall transcend his Ableton Live user situation and step out of the beaten path to create a multidimensional audiovisual experience.

3 Alumni

3.1 Former members

Research and Developpement:

- **Gabriel Downs** has a Bachelors in Honours Mathematics and Computer Science from McGill University (2019), where he did research in Artificial Intelligence and Computer Vision. He is interested in Free and Open Source Software, DJing, and music performance.
- **Marc Lavallée** possède une maîtrise en communication multimédia de l'Université du Québec à Montréal. Il a étudié l'ingénierie sonore à l'Université McGill et l'ingénierie électrique à l'école Polytechnique de Montréal. Il s'intéresse aux médias électroniques, aux logiciels libres et à l'art sonore.
- **Marie-Eve Dumas** possède un master en ingénierie mécanique (2017) ainsi qu'un un baccalauréat en mathématiques (2013) de l'Université McGill. Elle aime l'informatique graphique, les mathématiques et le logiciel libre.
- **Patrick Dupuis** possède un DESS en musiques numériques de l'Université de Montréal (2017). Il se passionne pour l'Open Source sous Linux, l'audio numérique et la synthèse sonore.
- **Harish Venkatesan** has a Master's degree in Music Technology from McGill University (2020) and a Bachelor's in Telecommunication Engineering from Ramiah Institiute of Technology (2017). His interests lie in audio DSP, room acoustic modeling and 3D audio.

Director of Strategic Partnership and Innovation Valorization at [SAT]:

- **Cédrick Lalaizon** possède un diplôme universitaire de technologie en techniques de commercialisation de l'Université René Descartes (Paris V), un diplôme de l'école de commerce EMLV et un master en marketing des services de l'INSEEC Paris. Passionné du XVIIIe siècle et admiratif du stoïcisme, il est aussi amateur de pâtisserie bretonne. *Directeur partenariat stratégique et valorisation des innovations de la [SAT]*

3.2 Former interns

3.2.1 2022

- **Audrey Doualot** is a Ph.D. in psychology and lecturer in the psychology department of the University of Quebec in Montreal and postdoctoral researcher in the art history department. Audrey's work on perception focuses on the interactions between humans and their living and built environment from a multisensory perspective.

3.2.2 2021

- **Elisabeth Fagnan** est étudiante au baccalauréat en ingénierie logiciel à Polytechnique Montréal. Elle s'intéresse aux applications des technologies multimédias dans le domaine des arts.
- **Hector Teyssier** est un étudiant complétant un baccalauréat en informatique et musique numérique à l'Université McGill. En plus de sa formation en informatique, il est très intéressé par la production musicale, que ce soit pour des morceaux électroniques ou bien pour des oeuvres qui mêlent l'audio à l'art visuel.
- **Jean Philippe Robitaille-Larratt** est un étudiant au baccalauréat en informatique et génie logiciel à l'Université du Québec à Montréal. Il s'intéresse aux gens et à tous leurs points de vue. L'ouverture et l'exploration sont les valeurs qu'il aimerait affirmer.
- **Corentin Gilbert** est étudiant en fin de parcours d'un double diplôme architecte-ingénieur coordonné par l'ENSA et l'École Centrale de Nantes. Explorateur éclectique des réalités altérées, il façonne leurs émergences par la narration, le dessin et le prototypage. Adeptes de performances artistiques, il se sent proche du libre et des Creative Commons.
- **Frédéric Lestage** possède un diplôme collégial en multimédia (2016) et débute sa quatrième année d'études en génie logiciel à l'école de technologie supérieure (ÉTS). Il s'intéresse aux arts, aux mathématiques et aux applications de la programmation dans le multimédia.

- **Io Andes Daza-Dillon** est un futur ingénieur logiciel de l'École de technologie supérieure (2021). Il possède un Baccalauréat ès arts de l'Université de Montréal (2014). Son ambition est de faire des choses intéressantes.

3.3 Interns' publications

- [Intrn1] Audrey Doualot. *La réalité virtuelle immersive & la transmission du sensible en architecture*. Post-doctorat — Université du Québec à Montréal. Jan. 2022.
- [Intrn2] Giulia Taurino. *Adaptive navigation in structured archives of immersive content*. Post-doctorat — Université de Montréal et metaLAB de Harvard. Sept. 2022.
- [Intrn3] Io Andes Daza-Dillon. *Modification du système de rendu audio de Mozilla Hubs*. Baccalauréat en génie logiciel — École de Technologie Supérieure (ÉTS). Apr. 2021.
- [Intrn4] Élisabeth Fagnan. *Conception d'expérience utilisateur et intégration pour application web dédiée à la spatialisation du son*. Baccalauréat en génie logiciel — Polytechnique Montréal. Aug. 2021.
- [Intrn5] Corentin Gilbert. *Déploiement d'espaces immersifs — design et conception*. ENSA Nantes et Ecole Centrale Nantes, architecture. Apr. 2021.
- [Intrn6] Frédéric Lestage. *Génération automatique de coordonnées de texture pour le projection mapping*. Baccalauréat en génie logiciel — École de Technologie Supérieure (ÉTS). Apr. 2021.
- [Intrn7] Jean Philippe Robitaille-Larratt. *Stratégie pour une communauté informatique ouverte, diverse et inclusive*. Informatique et génie logiciel — Université du Québec à Montréal. Sept. 2021.
- [Intrn8] Hector Teyssier. *Interpolation de réponse impulsionnelles en direct*. Bachelor (B-Sc), Université McGill. Aug. 2021.
- [Intrn9] Jack Atherton. *Collective VR Environment Creation by Example (canceled due to COVID-19)*. Ph.D. — CCRMA, Stanford University. Apr. 2020.
- [Intrn10] Mouna Belaid. *Conception d'architecture et prototypage de rendu immersif audiovisuel en infonuagique*. Baccalauréat en génie logiciel, Polytechnique Montréal. Aug. 2020.
- [Intrn11] Pierre Gruner. *Études et amélioration des stratégies de gestion et de diffusion des logiciels libres d'une organisation — cas d'études de la Société des Arts Technologiques*. Maîtrise Génie des Systèmes (Gestion des affaires), École de Technologie Supérieure (ÉTS). Aug. 2020.
- [Intrn12] Benjamin Langlois. *Étude et prototypage de hauteur-parleurs virtuels à l'aide d'un groupe de haut-parleurs sphériques*. Génie Électrique T3, Université de Sherbrooke. Apr. 2020.

- [Intrn13] Youness Salame. *Étude utilisateur de l'apprentissage de l'histoire en immersion*. Master en Science (bourse MITACS), HEC Montréal. Co-encadrée avec P.-M. Léger. Sept. 2020.
- [Intrn14] Maÿlis Merveilleux du Vignaux. *Étude utilisateur de l'apprentissage de l'histoire en immersion*. Master en Science (bourse MITACS), HEC Montréal. Co-encadrée avec P.-M. Léger. Sept. 2020.
- [Intrn15] Hantz-Carly Fleurant Vius. *Étude et prototypage d'interopérabilité de technologies de vidéoconférence web avec le logiciel de téléprésence Scenic*. Génie Informatique T2, Université de Sherbrooke. Apr. 2020.
- [Intrn16] Joseph Battesti. *Mesure et Visualisation de Réponse Impulsionnelle Directionnelle*. Génie Physique, Polytechnique Montréal. Aug. 2019.
- [Intrn17] Vincent Cusson. *Conversion d'oeuvres audio spatialisées vers un format standard*. Baccalauréat en communication profil médias interactifs, Université du Québec à Montréal (UQAM). Apr. 2019.
- [Intrn18] Léa Nugue. *Prototypage in situ d'expérience immersive*. Diplôme Supérieur de Recherche en Art (DSRA), École Supérieure d'Art Annecy Alpes (ESA AA). Sept. 2019.
- [Intrn19] Jérôme Berthelin. *Projection vidéo sur objets mobiles*. Diplôme d'ingénieur, École nationale supérieure d'informatique pour l'industrie et l'entreprise (ENSIIE) Paris-Évry. Apr. 2018.
- [Intrn20] Miriana Couvret-Michel. *Prototypage de Scénographie Immersive*. Master Création Numérique : Interactivité, Générativité, Université Toulouse II Jean Jaurès. May 2018.
- [Intrn21] Julien Wantz. *Synthèse temps réel de réponse impulsionnelle pour la simulation d'acoustique*. Diplôme d'ingénieur, École Centrale de Nantes. Nov. 2018.

4 Hardware prototypes from the Metalab

4.1 Audiodice



The first prototype of the audio spatialization system *Audiodice* is composed of five boxes each containing twelve individually controllable speakers. This system was imagined with the aim of populating the physical space with sound sources, where the usual systems diffuse sound from the periphery.

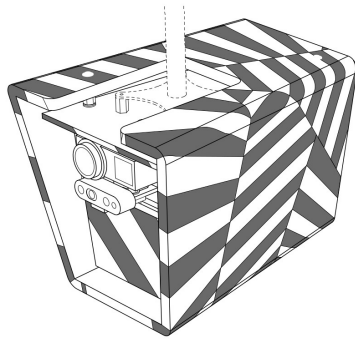
In its current form, *Audiodice* has 60 audio channels, and can be configured and controlled to broadcast an immersive audio environment through the use of SATIE. It can also be used with other audio tools, whether spatialized or not.

The use cases of such a system are numerous and to be explored, here are those which hold our attention :

- Represent a specific sound source (musician, singer, etc.) by an omnidirectional physical source
- Augment a traditional audio system (such as an audio dome) with spatialized sources placed in the listening space
- Enable the acoustic characterization of physical spaces by measuring impulse responses

Demonstration “Audiodice - Multidirectional Speakers for Immersive Spaces : <https://vimeo.com/519938720>

4.2 Mirador



*Mirador*¹ is a mobile device under development, whose objective is to facilitate the deployment of temporary immersive video spaces. In a reduced format, it is composed of a video projector and a color and infrared camera (calibrated in the manner of a stereoscopic couple), as well as an embedded computer. For the software part, *Mirador* currently relies on *Splash* (5.6), *Calimiro* (5.7) and *LivePose* (5.8).

In practice, *Mirador* will be able to measure the geometry of the projection surface it faces and adapt the video content to that surface. It is also planned to be able to run multiple devices collaboratively to cover a larger area.

The video content comes from the device itself or from video sources on the local network, taking advantage of *Switcher* (5.4) or the Network Device Interface (NDI)² audio-visual streaming tool, widely adopted in the industry.

Demonstration: <https://vimeo.com/549423960>

¹<https://gitlab.com/sat-metalab/hardware/mirador>

²<https://ndi.tv>

5 Software from the Metalab

5.1 SATIE



The Spatial Audio Toolkit for Immersive Environments (*SATIE*) is an audio spatialization engine capable of live rendering of dense (several hundred sound sources) and dynamic (individual sound source creation, removal and control) audio scenes. *SATIE* uses the SuperCollider language and the OSC protocol, allowing control from external tools such as 3D engines and programming languages.

SATIE facilitates the prototyping of innovative immersive sound spaces. Its features include:

- Parameterizable spatializer for specific sound devices, such as those of the SAT (mini domes, dome, cubes, stereo)
- Capable of live and simultaneous rendering for multiple types of speaker systems
- OSC control from 3D engines or other
- Ability to write audio source extensions, audio effects, parameter settings and spatializers
- Playback of audio files
- Audio synthesis
- Tested under Linux and OSX, should work under Windows
- Distributed as a quark SuperCollider

Development of *SATIE* began in 2015. It is distributed under the GPL and can be downloaded here: <https://gitlab.com/sat-metalab/satie>.

Demonstrations:

- Using SATIE with Mozilla Hubs: <https://vimeo.com/549433773>
- Using SATIE with two spatialization devices, the Audiodice and the Satosphere. (Prototyping a) Immersive Navigation Experience in the Montreal Symphony Orchestra: <https://vimeo.com/519940468>
- Generation of a 3D animation with Blender and SATIE: <https://vimeo.com/398346954>
- Real time navigation in a 3D environment (EiS + SATIE) : <https://vimeo.com/265607999>

5.2 vaRays



VaRays, for *virtual audio rays*, is a tool allowing the calculation in real time of impulse responses of a virtual environment. It takes into account the geometry of the environment, the position of the listener and that of the sound objects. This tool is in its early stages of development, having started in 2018. It is currently used in conjunction with *EiS* (5.3), for geometry, and *SATIE* (5.1), for impulse response based reverberation rendering.

VaRays is distributed under LGPL license and can be downloaded here: <https://gitlab.com/sat-metalab/vaRays>.

Demonstrations:

- Live sound ray launching, and Auralization, of 3D audio scene with vaRays – Presentation at the SMC conference: <https://vimeo.com/574450746>
- Navigation in the model and sounds of Bretez's project (Paris in the 18th century) : <https://vimeo.com/507255065>
- Live acoustic simulation with vaRayz : <https://vimeo.com/manage/videos/484122810>

5.3 Édition in Situ



Edition in Situ (EiS) is an immersive environment creation platform designed to allow the prototyping of 3D scenography. The hypothesis behind the development of this platform is that by allowing creation in an immersive context, the user will have the ability to experiment in more detail and more quickly, leading to a shorter creation cycle and original content. It integrates existing tools (*Splash* (5.6), *SATIE* (5.1)) as well as new software to allow the user to manipulate the 3D scene using devices adapted to virtual reality interaction (currently, HTC Vive controllers¹).

Its features includes:

- Real-time rendering for specific immersive space (format *cubemap*²)
- Navigation in the 3D space;
- Importing content (objects, textures, sounds);
- Transformation of objects (translation, rotation, scale) ;
- Animation ;
- Sound spatialization with *SATIE*.

The development of *EiS* started in 2015. It is distributed under GPL license and can be downloaded here: <https://gitlab.com/sat-metalab/EditionInSitu>.

Demonstrations:

- Basic operations to create an immersive installation from scratch with EiS : <https://vimeo.com/339326874>
- Experimentation with EiS - History of SAT : <https://sat.qc.ca/fr/nouvelles/presentation-experimentale-de-la-plateforme-eis>
- Fulldome animation drafting : <https://vimeo.com/226298383>

¹<https://www.vive.com/us/>

²https://en.wikipedia.org/wiki/Cube_mapping

5.4 Switcher



Switcher is a tool for creating collaborative experiences over the Internet. It is capable of integrating multiple technologies, such as network protocols and other applications that operate in real time. It is most often used for its ability to simultaneously transmit multi-channel audio, video and low latency data streams over the Internet.

Switcher is actually more generic and relies on an architecture of services (called quiddities). Each quiddity can be created and deleted dynamically, but can also be controlled using property read/write or method invocation. Quiddity monitoring is done using queries to its information tree, possibly through serialized string in the JSON format³. *Switcher* provides introspection mechanisms allowing it to be used as a library for developing higher-level applications (in C++, NodeJS⁴ or Python). *Switcher* also allows to save and download the state of the active quiddities at a given time.

Most quiddities produce and/or consume data streams using the *shmdata* library (see section (5.5)), thus allowing the transfer of streams between other quiddities and/or applications. For example, a webcam service (*v4l2src* service) provides a video stream that can simultaneously connect to several other services, such as a window display (*glfwin* service), a file recorder (*avrec* service), a low latency network transmitter (*SIP* service), but also simultaneously to other software compatible with *shmdata*.

The development of *Switcher* started in 2012. It is distributed under GPL and LGPL licenses. It can be downloaded here: <https://gitlab.com/sat-metalab/switcher>

Switcher is the basis for more specific telepresence systems, also developed at SAT:

- Scenic, a web-based graphical interface to switcher: <https://sat-mtl.gitlab.io/telepresence/scenic>
- Bibliolab, networked connection of Montreal libraries: <https://sat.qc.ca/en/scenic-telepresence>
- Scènes ouvertes, a network of about twenty Quebec venues: <https://sat.qc.ca/en/scenic-telepresence>

Demonstrations:

³<https://www.json.org/json-en.html>

⁴<https://nodejs.org/en/>

- The Bluff theater project: <https://vimeo.com/530865445>
- Streaming test from a fisheye camera to two fulldomes : <https://vimeo.com/266687560>
- Waterfall Music during Linux Audio Conference 2013: <https://vimeo.com/304831248>
- The Drawing Space - Siggraph 2014: <https://vimeo.com/304826675>

5.5 Shmdata



Shmdata is a tool allowing zero-copy transmission between programs running on the same computer. *Shmdata* works without a server: an application emitting a data stream (a *shmdata*) must provide a path (e.g. */tmp/myshm*) which will be used by the programs wishing to read the stream. *Shmdata* allows the transmission of data between programs with very low latency.

The communication paradigm of *shmdata* is *1-to-multiple*, i.e. the writer of a stream makes the images of the stream available to possibly several followers (readers). During a transmission, the subscribers and the writer are able to leave the transmission and return along the way, without having to restart the communication. A *shmdata* can be resized during transmission. In addition, *shmdata* are typed, using a string published by the writer at each (re)connection.

Shmdata can be used by applications written in C++ or Python 3; a GStreamer⁵ element is also provided.

The development of *shmdata* started in 2012. It is distributed under LGPL license. It can be downloaded here: <https://gitlab.com/sat-metalab/shmdata>

5.6 Splash



Splash is a real-time projection mapping engine capable of handling a theoretically unlimited number of projectors. It has been designed as a generic and modular tool, allowing it to be used for any projection surface geometry. It is used in production at SAT in the Satosphere, which is equipped

⁵<https://gstreamer.freedesktop.org/>

with eight HD projectors. *Splash* is compatible with *shmdata* (5.5) for the reception of the video stream to be broadcast and for the communication in real time of 3D model from Blender⁶.

Characteristics:

- number of projectors unlimited (except by the power of the computer and the number of video outputs)
- support of numerous video codecs, including Hap⁷
- support for Video4Linux2 compatible capture cards
- semi-automatic calibration of projectors (the user must have a 3D model of the projection surface)
- projection deformation (*warping*)
- automatic calculation of the overlapping zones (blending)
- automatic colorimetric calibration of the projectors
- Python 3 compatible

The development of *Splash* started in 2014. It is distributed under GPL license and can be downloaded here : <https://gitlab.com/sat-metalab/splash>.

Demonstrations:

- Remote rendering of a multi-output OpenGL software : <https://vimeo.com/456217092>
- Calibration of projection mapping with *Splash* : <https://vimeo.com/314519230>
- Projection on mobile objects : <https://vimeo.com/268028595>

5.7 Calimiro

⁶<https://blender.org>

⁷<https://github.com/Vidvox/hap>



Calimiro is a tool for the calibration of immersive environment with structured light. It takes into account multi-projector environments. This tool is in its early stages of development, having started in 2018. It is used in conjunction with *Splash* (see section 5.6).

This software library is distributed under LGPL license and can be downloaded here : <https://gitlab.com/sat-metalab/calimiro>

Demonstrations:

- Automated generation of UV coordinates: <https://vimeo.com/544570900>
- Automatic calibration of video projection with Calimiro and Splash : <https://vimeo.com/499279721>
- Automatic projection calibration: <https://vimeo.com/329425110>

5.8 LivePose

LivePose is a command line tool for capturing human poses and movements in real time, using several popular computer vision models. Wishing to focus on the sustainability of its uses, we chose to reimplement these models based on the deep learning module of OpenCV⁸ rather than using the implementations made in the research work of the different laboratories. As OpenCV is available on many hardware platforms, and benefits from the support of a very strong community, we expect less maintenance of each model.

Currently, it is possible to choose between the pose estimation algorithms OpenPose⁹ and PoseNet¹⁰, depending on whether one wishes to optimize for accuracy or speed. *LivePose* also uses custom filters to extract additional information from the poses, such as gesture detection: hand salutes, jumps, etc.

Another specificity of *LivePose* is to be able to run on embedded hardware. In particular, *LivePose* works on NVIDIA Jetson. One of the objectives of the tool is to allow the deployment of a network of cameras to allow the capture of poses of many users on a large area.

⁸<https://opencv.org>

⁹<https://github.com/CMU-Perceptual-Computing-Lab/openpose>

¹⁰<https://github.com/google-coral/project-posenet>

LivePose is designed to be used in conjunction with 3D engines such as Mozilla Hubs, or immersive environments such as the Satosphere; in this case, filters can trigger actions based on the gestures of the participants, thus constituting a user interface that is based on movements and gestures, freeing itself from manual input.

This software library is distributed under LGPL license and can be downloaded here: <https://gitlab.com/sat-metalab/livepose>.

6 News Feeds

Several videos are available on the Metalab accounts:

- The videos of the Metalab technical demonstrations:
<https://vimeo.com/satmetalab>
- Some videos of works in 360 format:
<https://www.youtube.com/channel/UCqu0L3uSt0BmZ3mfaHkwrFw>

More info on the Metalab's work can be found on the [SAT] website:

- A blog communicating some of Metalab's achievements :
<http://sat.qc.ca/fr/recherche/metalab>
- A list of scientific publications:
<https://sat.qc.ca/fr/recherche/publications-du-metalab>

7 Scientific communications

7.1 Articles

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- [Pres28] Maïlis Merveilleux-du-Vignaux and Youness Salame. “Le projet Serdaigle : mesure de l’expérience utilisateur pour l’apprentissage avec des dispositifs immersifs”. In: *Émission Avez-vous du WIFI ? de la radio CISM (89,3)*. Montréal, June 2019.
- [Pres29] Julien Puget et al. “Rapid Prototyping of Immersive Video for Popularization of Historical Knowledge”. In: *13th International Conference on Tangible, Embedded, and Embodied Interaction (ACM TEI)*. Tempe, Arizona, USA, Mar. 2019.
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- [Pres31] Michał Seta. “La spatialisation du son avec SATIE”. In: *Initiation à la création sonore immersive, formation donnée par Mourad Bennacer*. [SAT], Montréal, Apr. 2019.
- [Pres32] Michał Seta and Emmanuel Durand. “Démonstration des travaux en son du Metalab”. In: *Sounds in the City Workshop*. CIRMMT, Montréal, June 2019.

- [Pres33] Zack Settel. “6DoF Navigation of Ambisonics Sound Field Sources: Initial Explorations”. In: *127e Convention de l’Audio Engineering Society (AES)*. New York, Oct. 2019.
- [Pres34] Nicolas Bouillot. “Initiation à la téléprésence dans les projets artistiques (participation en tant qu’expert)”. In: *Conférence internationale des arts de la scène (CINARS)*. Nov. 2018.
- [Pres35] Nicolas Bouillot. “Le Metalab : le laboratoire de recherche de la [SAT]”. In: *Maîtrise en création numérique à l’UQAT*. Rouyn-Noranda, Dec. 2018.
- [Pres36] Nicolas Bouillot. “Le Metalab : le laboratoire de recherche de la [SAT]”. In: *Introduction à la téléprésence artistique pour les artistes (SAT/ENTC), formation donnée par Julien Brun*. Montréal, Dec. 2018.

8 Artworks using Metalab tools

8.1 Oeuvres utilisant les outils du Metalab

- [Art1] Michał Seta. *Mimoidalnaube*. 2020. url: <http://djamnot.xyz/pages/works/mimoidalnaube.html> (visited on 01/13/2022).
- [Art2] Michał Seta. *Fadeferra*. 2019. url: <http://djamnot.xyz/pages/works/Fadeferra.html> (visited on 01/13/2022).
- [Art3] Zack Settel. *Aquakhoria*. 2016. url: <http://aquakhoria.com/> (visited on 01/13/2022).
- [Art4] Nicolas Bouillot. *Waterfall Music*. 2013. url: <https://sat.qc.ca/fr/evenements/waterfall-music> (visited on 01/13/2022).
- [Art5] Julien Brun and Vincent de Repentigny. *Dieu Est Un DJ*. 2012. url: <https://sat.qc.ca/fr/evenements/dieu-est-un-dj> (visited on 01/13/2022).
- [Art6] Intact collective. *Just in Case*. 2011. url: <http://www.intact01.net/emflaboral-hangarsat/> (visited on 01/13/2022).
- [Art7] Valérie Cordy, Laurence Moletta, Alexandre Quessy, and Michał Seta. *Soupe Transatlantique*. 2011. url: <https://sat.qc.ca/fr/evenements/soupe-transatlantique> (visited on 01/13/2022).
- [Art8] Warne Paul and Mike Wozniowski. *Breaking The Ice*. 2010. url: <https://sat.qc.ca/fr/evenements/briser-la-glace> (visited on 01/13/2022).

9 History

- **2002** Création du Metalab
- **2003 - 2007** Programme de recherche TOT [Territoires Ouverts - Open Territories] sur les nouveaux médias, ayant pour objectifs de permettre aux artistes d'explorer la création en ligne sur les réseaux IP à large bande de la recherche et de les doter d'applications adaptées au domaine des nouveaux médias. Ainsi, les chercheurs se sont basés sur quatre axes de recherches : l'immersion, la webdiffusion, la téléprésence et la télévisite. Naissance des logiciels LighTWIST pour la correction de perspectives avec projections multiples sur surfaces variables ; AudioTWIST pour la création d'un système d'environnement audio 3D ; pixelTANGO pour le mixage vidéo numérique ; nSLAM pour la création d'une application audio multicanale ; ainsi que des stations de téléprésence fonctionnant sous l'application TéléCHACHA.

Le projet TOT [Territoires Ouverts - Open Territories] a bénéficié du soutien financier du ministère du Patrimoine canadien par l'entremise du Programme de culture canadienne en ligne, ainsi que de celui du gouvernement du Québec par l'entremise des ministères de la Culture et des Communications, des Affaires municipales et des Régions et du Développement économique, de l'Innovation et de l'Exportation.

- **2005** Création du département du Labodome, dont le mandat est de développer des techniques de production et de diffusion pour les environnements immersifs, principalement basé sur des expériences immersives de groupe sous des dômes immersifs
- **2006** Création du département SAT[TRANSFORM], soutenant le volet éducatif de la Société des arts technologiques [SAT], dont la vocation est la transmission de connaissances issues des activités artistiques et de recherche de la SAT. En 2011, SAT[TRANSFORM] est devenu Campus.

- **2008** Six ans après le lancement du programme Territoires Ouverts (recherche et expérimentation artistique avec des dispositifs immersifs), la SAT inaugure en 2008, pour le 400^e de Québec, son premier dôme immersif gonflable lors de Domagaya.
- **2008-2014** Programme d'échanges croisés Contamine. Échanges artistiques nationaux et internationaux ayant pour objectif de réaliser des résidences croisées impliquant la recherche et la création multidisciplinaire et téléprésentielle.
- **2010** Reconnaissance de la SAT par l'ENoLL (European Network of Living Labs) en 2010, comme un laboratoire vivant qui favorise la cocréation, l'innovation ouverte et le partage de connaissances.
- **2011** Inauguration de la Satosphère, le dôme immersif de la SAT dont la conception est issue des recherches du Metalab
- **2012-2015** Programme de recherche en Scénographie immersive, Interaction Intelligente et expérience réseau, soutenu par le "Programme de soutien à la recherche. Volet 1: Soutien à des projets de recherche ou à des organismes et regroupements stratégiques" du Ministère de l'Économie, de l'Innovation et de l'Exportation (MEIE)
- **2015-2017** Programme de recherche " Développement et usage d'une plateforme réseau de cocréation, d'innovation ouverte et de partage de connaissances ", supporté par le Ministère de l'Économie, de l'Innovation et de l'Exportation (MEIE) du Québec. Un des grands objectifs de la plateforme est de faciliter la communication, la formation, la concertation, la prise de décision et la coordination de projet complexes pour les équipes qui travaillent en réseau. Pour ce faire elle mise sur des approches qui reproduisent, dans les espaces physiques hybrides (EPH) ou les espaces virtuels partagés (EVP), les codes comportementaux usuels qui font usage de la communication non verbale (reconnaissance des interlocuteurs, posture, contact du regard, positionnement et mouvements dans l'espace, etc.)
- **2016-2018** Bibliolab est un projet qui vise à connecter les bibliothèques de la ville de Montréal pour favoriser les rencontres et interactions entre les citoyens. L'implantation d'un dispositif immersif de téléprésence permet aux usagers distants de partager simultanément des expériences culturelles, pédagogiques, et collaboratives. Le projet bénéficie d'une aide financière gouvernementale provenant du Secrétariat à la région métropolitaine du ministère des Affaires municipales et de l'Occupation du territoire par l'entremise du Fonds d'initiative et de ray-

onnement de la métropole. Il bénéficie également d'une aide financière de la Caisse de la Culture Desjardins et de la ville de Montréal.

- **2017** Le Metalab connecte 19 lieux de diffusion en inaugurant Scènes ouvertes, un réseau de salles de spectacle connectées à travers le Québec. Aboutissement de plusieurs années de recherche, la connexion est rendue possible grâce à un nouveau dispositif de téléprésence scénique. Il permettra la mise en œuvre de projets collaboratifs d'envergure et le développement d'un nouveau langage scénographique. Le réseau Scènes ouvertes est financé par le Ministère de la Culture et des Communications (MCC) dans le cadre du plan culturel numérique du Québec et d'autre part par Patrimoine Canada.
- **2017** Création du département de valorisation de la recherche de la SAT. Son premier mandat consiste à prendre en charge le réseau Scènes Ouvertes.
- **2017 - 2019** Programme de recherche "Création d'outils et de procédés pour la création et l'édition in situ d'espaces immersifs interactifs", supporté par le Ministère de l'Économie et de l'Innovation (MÉI) du Québec. Bâti sur l'observation que la création de contenu pour les espaces immersifs est fortement influencée par ceux-ci et nécessite donc soit une connaissance préalable de l'espace cible, soit la possibilité d'expérimenter pendant le processus créatif, ce programme de recherche s'est concentré sur la mise en place de méthodes et d'outils pour permettre le prototypage de contenu immersif en situation d'immersion (in situ), et ce quelque soit l'espace immersif.
- **2018 - 2019** Réalisation du projet "Bancs d'essai en téléprésence pour le réseau Scènes ouvertes". Ce projet, financé par le ministère de la Culture et des Communications (MCC), vise à accélérer l'expérimentation en téléprésence scénique, de collecter les apprentissages acquis au cours de ces expérimentations et de les partager à l'ensemble du réseau de salles connectées. Le projet initie 10 projets en téléprésence.
- **2019 - 2022** Programme de recherche "Immersive Collaborative, Simultanée et Adaptative", soutenu par le MÉI. L'objet de ce programme est de favoriser le développement de l'utilisation de l'immersion dans des contextes variés, en assouplissant les contraintes techniques qui limite leur déploiement. À plus long terme il s'agit de permettre à l'immersion d'être intégré à tout type de lieu, pour en favoriser l'utilisation.
- **2019 - 2022** Programme de recherche "Simulation Acoustique Volumétrique en VR/AR/XR", soutenu par le MÉI. Ce programme a pour ambition l'étude et la mise en œuvre d'une méthodologie complète de simulation d'acoustique pour

les espaces immersifs. Ces recherches couvrent la capture sonore volumétrique, le rendu acoustique directionnel et l'augmentation sensorielle des lieux dédiés à l'écoute de sons spatialisés.

- **2020** Création du département Satellite, le nouveau hub virtuel transdisciplinaire de la SAT, offrant une plateforme de diffusion en ligne et encourageant le développement d'expériences immersives hybrides entre participants distants dans un monde virtuel et groupes de participants dans des salles de diffusion.
- **2020** La SAT devient membre de QuébecInnove, à titre de membre issu de la recherche
- **2021- 2023** Le projet "Scenic Light 5G", soutenu par le programme de soutien aux organismes de recherche et d'innovation, volet 2 du MÉR. Développement d'un dispositif léger et mobile pour la scénographie téléprésentielle sur réseau 5G.
- **2021-2022** Le projet "Stratégie de pré commercialisation du plancher haptique", soutenu par le programme de soutien aux organismes de recherche et d'innovation, volet 2 du MÉR. Développement d'un plancher haptique constitué d'un maillage de dalles mobiles, contrôlées de façon indépendante entre elles. Ce dispositif développé entièrement au Québec permet de faire ressentir des vibrations et mouvements aux usagers dans des lieux aux dimensions variables.

10 Software code contribution statistics

Code additions since 2012

	# commits	# contributors	# versions
EditionInSitu	656	11	0
calimiro	183	3	4
doc-satie	64	5	0
doc-shmdata	64	3	0
doc-splash	118	3	0
ndi2shmdata	39	3	5
poire	150	8	0
pysatie	162	4	1
satie	1,143	9	14
satie4blender	173	4	2
satie4unity	126	1	1
satie4unityExample	219	2	2
shmdata	698	12	91
splash	2,121	9	78
livepose	187	5	0
switcher	3,260	21	138
tools-ambir	163	9	0
varays	375	9	8

Active contributors per year

	11/12	12/13	13/14	14/15	15/16	16/17	17/18	18/19	19/20	20/21	21/22
EditionInSitu	0	0	0	0	0	0	5	4	7	7	2
calimiro	0	0	0	0	0	0	0	2	2	3	2
doc-satie	0	0	0	0	0	0	0	0	0	1	5
doc-shmdata	0	0	0	0	0	0	0	0	0	1	3
doc-splash	0	0	0	0	0	0	0	0	0	1	3
ndi2shmdata	0	0	0	0	0	0	0	1	2	1	2
poire	0	0	0	0	0	0	0	0	0	5	4
pysatie	0	0	0	0	0	0	1	1	1	4	2
satie	0	0	0	0	0	0	3	7	5	5	1
satie4blender	0	0	0	0	0	3	2	0	1	2	1
satie4unity	0	0	0	0	0	0	1	1	1	1	0
satie4unityExample	0	0	0	0	0	0	1	1	1	2	0
shmdata	2	4	5	2	2	4	2	3	4	4	3
splash	0	0	1	2	2	2	4	3	2	3	2
livepose	0	0	0	0	0	0	0	0	2	4	4
switcher	0	3	5	3	6	6	3	8	6	6	5
tools-ambir	0	0	0	0	0	0	0	0	4	4	4
varays	0	0	0	0	0	0	0	4	4	6	1

Number of commits per year

	11/12	12/13	13/14	14/15	15/16	16/17	17/18	18/19	19/20	20/21	21/22
EditionInSitu	0	0	0	0	0	0	169	108	260	114	3
calimiro	0	0	0	0	0	0	0	89	34	38	22
doc-satie	0	0	0	0	0	0	0	0	0	39	20
doc-shmdata	0	0	0	0	0	0	0	0	0	48	16
doc-splash	0	0	0	0	0	0	0	0	0	57	61
ndi2shmdata	0	0	0	0	0	0	0	14	16	3	5
poire	0	0	0	0	0	0	0	0	0	77	73
pysatie	0	0	0	0	0	0	30	7	26	67	31
satie	0	0	0	0	0	0	265	418	251	197	11
satie4blender	0	0	0	0	0	10	68	0	20	63	12
satie4unity	0	0	0	0	0	0	31	8	30	57	0
satie4unityExample	0	0	0	0	0	0	116	32	11	60	0
shmdata	65	129	59	60	175	62	4	30	21	71	19
splash	0	0	341	353	421	347	182	132	111	155	71
livepose	0	0	0	0	0	0	0	0	15	119	52
switcher	0	377	585	569	474	438	85	141	230	293	57
tools-ambir	0	0	0	0	0	0	0	0	75	20	68
varays	0	0	0	0	0	0	0	119	73	175	7

Number of releases per year

	11/12	12/13	13/14	14/15	15/16	16/17	17/18	18/19	19/20	20/21	21/22
EditionInSitu	0	0	0	0	0	0	0	0	0	0	0
calimiro	0	0	0	0	0	0	0	0	0	4	0
doc-satie	0	0	0	0	0	0	0	0	0	0	0
doc-shmdata	0	0	0	0	0	0	0	0	0	0	0
doc-splash	0	0	0	0	0	0	0	0	0	0	0
ndi2shmdata	0	0	0	0	0	0	0	1	2	1	1
poire	0	0	0	0	0	0	0	0	0	0	0
pysatie	0	0	0	0	0	0	0	0	0	1	0
satie	0	0	0	0	0	0	2	7	2	3	0
satie4blender	0	0	0	0	0	0	0	0	0	1	1
satie4unity	0	0	0	0	0	0	1	0	0	0	0
satie4unityExample	0	0	0	0	0	0	2	0	0	0	0
shmdata	2	6	8	1	3	19	0	9	4	29	10
splash	0	0	0	6	7	15	12	4	2	20	12
livepose	0	0	0	0	0	0	0	0	0	0	0
switcher	0	0	7	2	4	43	14	6	8	39	13
tools-ambir	0	0	0	0	0	0	0	0	0	0	0
varays	0	0	0	0	0	0	0	1	0	7	0